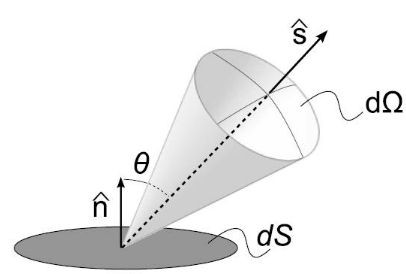
Light propagation models

The radiance or "specific intensity"

The radiated energy per unit time, unit *projected* area, and unit solid angle:

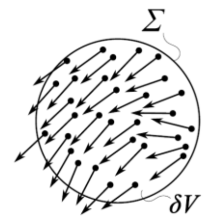
$$dtdS\cos\theta \mathrm{d}\Omega I(\mathbf{r},\hat{\mathbf{s}})$$

The radiance $I(\mathbf{r},\hat{\mathbf{s}})$ defines how much intensity radiates in a certain direction



Relation to Poynting's vector

- Radiance is best understood in the context of ray optics
- In terms of the Poynting's vector, it may be understood as the "average" over a certain volume or time



S=$|<\mathbf{S}>|$ $\qquad$ $I(r,\hat{\mathbf{s}}) = \int_0^\infty Sw(S,\theta,\varphi)S^2 dS$

$w(S,\theta,\varphi)dS_x dS_y dS_z$ – The "probability" of S having a certain value and direction

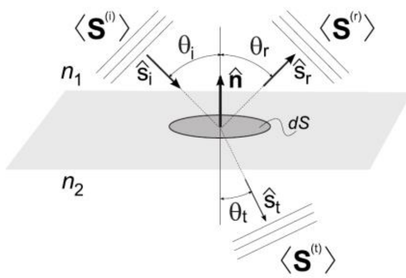
Relation to Poynting's vector

$<\mathbf{S}>$ and $I(r,\hat{\mathbf{s}})$ have a different mathematical structure:

- $I(r,\hat{\mathbf{s}})$ is a function defined over 5D space
- $<\mathbf{S}>$ is a vector defined over 3D space

$$<\mathbf{S}> = A_1\hat{\mathbf{s}}_\mathbf{0} \quad \Longrightarrow \quad I(r,\hat{\mathbf{s}}) = A_2\delta(\hat{\mathbf{s}} - \hat{\mathbf{s}}_\mathbf{0})$$

Reflectivity and Transmissivity at surface boundaries



Snell's and Fresnel's equations apply, and need to be defined in terms of the Poynting vector or radiance

$$\langle \mathbf{S}^{(i)} \rangle = \frac{c}{8\pi} \Re\{\mathbf{E}^{(i)} \times \mathbf{H}^{(i)*}\} = \frac{c}{8\pi}\sqrt{\frac{\epsilon_1}{\mu_1}} |\mathbf{E}^{(i)}|^2 \hat{\mathbf{s}}_i$$

Reflectivity and Transmissivity at surface boundaries

Reflection and transmission depend on the polarization of light:

$$\langle \mathbf{S}^{(r)} \rangle_{\|} = \frac{c}{8\pi}\sqrt{\frac{\epsilon_1}{\mu_1}} |r_{\|}|^2 E_{\|}^2 \hat{\mathbf{s}}_r, \qquad \langle \mathbf{S}^{(t)} \rangle_{\|} = \frac{c}{8\pi}\sqrt{\frac{\epsilon_2}{\mu_2}} |t_{\|}|^2 E_{\|}^2 \hat{\mathbf{s}}_t,$$

$$\langle \mathbf{S}^{(r)} \rangle_{\perp} = \frac{c}{8\pi}\sqrt{\frac{\epsilon_1}{\mu_1}} |r_{\perp}|^2 E_{\perp}^2 \hat{\mathbf{s}}_r, \qquad \langle \mathbf{S}^{(t)} \rangle_{\perp} = \frac{c}{8\pi}\sqrt{\frac{\epsilon_2}{\mu_2}} |t_{\perp}|^2 E_{\perp}^2 \hat{\mathbf{s}}_t,$$

where $r_{\|,\perp}$ and $t_{\|,\perp}$ are the Fresnel formulas

Reflectivity and transmissivity at surface boundaries

If we assume that $\mu_1 = \mu_2$, and non-polarized light we obtain (after some derivation):

$$R = \frac{1}{2}\left[\left(\frac{n_1 cos\theta_t - n_2 cos\theta_i}{n_1 cos\theta_t + n_2 cos\theta_i}\right)^2 + \left(\frac{n_1 cos\theta_t - n_2 cos\theta_i}{n_1 cos\theta_t + n_2 cos\theta_i}\right)^2\right]$$

$$T = \frac{1}{2}\left[\left(\frac{n_1 cos\theta_i}{n_1 cos\theta_t + n_2 cos\theta_i}\right)^2 + \left(\frac{n_1 cos\theta_i}{n_1 cos\theta_t + n_2 cos\theta_i}\right)^2\right]$$

Fluence rate or "average intensity"

$|<\mathbf{S}>|/c$ represents the energy density of the EM field (in the far field)

$$\int_{\delta V} |<S>|/c\, dv \rightarrow \frac{1}{c}\int_{4\pi}\int_0^\infty Sw(S,\theta,\varphi)S^2 dS d\Omega \qquad \text{where} \quad S{=}|<\mathbf{S}>|$$

$$= \frac{1}{c}\int_{4\pi} I(r,\hat{\mathbf{s}})d\Omega$$

We define the fluence rate: $U(\mathbf{r}) = \int_{(4\pi)} I(\mathbf{r},\hat{\mathbf{s}})\mathrm{d}\Omega. \quad \left(Watts/cm^2\right)$

The energy density is given by: $u(\mathbf{r}) = \frac{U(\mathbf{r})}{c_0}. \quad \left(Joules/cm^3\right)$

The flux

$< \mathbf{S} >$ represents the flux of the EM field

$$\int_{\delta V} < \mathbf{S} > dv \rightarrow \frac{1}{c} \int_{4\pi} \int_{0}^{\infty} S\hat{s}w(S,\theta,\varphi)S^2 dS d\Omega = \int_{4\pi} I(r,\hat{\mathbf{s}})\hat{\mathbf{s}} d\Omega$$

The flux related to the radiance:

$$\mathbf{J}(\mathbf{r}) = \int_{(4\pi)} I(\mathbf{r},\hat{\mathbf{s}})\hat{\mathbf{s}}\mathrm{d}\Omega. \qquad \left(Watts/cm^2\right)$$

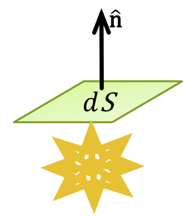
Radiance from the sun

$$I(r,\hat{\mathbf{s}}) = A\,\delta(\hat{\mathbf{s}} - \widehat{\mathbf{s}_0})$$

Light propagates in a single direction

Isotropic radiance

$$I(r,\hat{\mathbf{s}}) = \frac{P_0}{4\pi} \qquad \int_{4\pi} I(r,\hat{\mathbf{s}})d\Omega = P_0$$



Radiation through a surface

$$\int_{2\pi} I(r,\hat{\mathbf{s}})\hat{\mathbf{s}}\cdot\hat{\mathbf{n}}dS$$

The emitted power per unit area is dS

$I(r,\hat{\mathbf{s}})\hat{\mathbf{s}}\cdot\hat{\mathbf{n}} = \frac{P_0}{4\pi}cos\theta$

Lambertian surfaces

- A Lambertian surface is highly scattering and its radiance distribution is independent of how it is illuminated $I'(r,\hat{\mathbf{s}}) = A\,\hat{\mathbf{s}}\cdot\hat{\mathbf{n}} = A\cos(\theta)$

  Where $\hat{\mathbf{n}}$ is the surface normal

- Examples: walls, paper, plastics, and generally any "natural looking" white surface

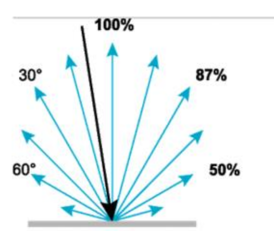
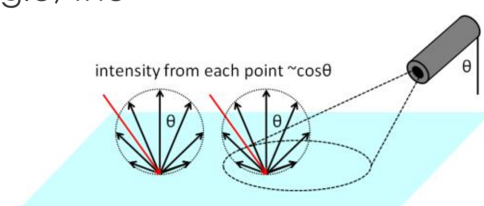


Fig. 6.5 Lambertian surface.

Lambertian surfaces

- Although the radiance $I'(r,\hat{\mathbf{s}})$ is angle-dependent, the perceived luminescence is still isotropic

- When increasing the angle, the radiace $I'(r,\hat{\mathbf{s}})$, decreases but the area that contributes to the illumination increases



Remember: Lambertian radiation is essentially isotropic

Conservation of power in terms of radiance (no scattering)

- Poynting's theorem (far field):

$$\frac{1}{c}\frac{\partial|<\mathbf{S}>|}{\partial t}+\nabla\cdot<\mathbf{S}>+\alpha<\mathbf{S}>=0$$

- Radiance equivalent: $$\frac{1}{c}\frac{I(r,\hat{\mathbf{s}})}{\partial t}+\nabla\cdot[I(r,\hat{\mathbf{s}})\hat{\mathbf{s}}]+\alpha I(r,\hat{\mathbf{s}})=0$$

- Or alternatively: $$\frac{1}{c}\frac{I(r,\hat{\mathbf{s}})}{\partial t}+\hat{\mathbf{s}}\cdot[\nabla I(r,\hat{\mathbf{s}})]+\alpha I(r,\hat{\mathbf{s}})=0$$

Loss and gain

- $I(r,\hat{\mathbf{s}})$ losses via absorption and scattering to other directions

The loss term: $(\mu_s + \mu_a)I(r,\hat{\mathbf{s}})=\mu_t I(r,\hat{\mathbf{s}})$

- $I(r,\hat{\mathbf{s}})$ gains from light from other directions scattered into its direction

The gain term:

$$\int_{4\pi} I(r,\widehat{\mathbf{s}'})\,|\mathrm{f}(\hat{\mathbf{s}},\hat{\mathbf{s}}_0)|^2\, d\Omega' = \mu_t \int_{4\pi} I(r,\widehat{\mathbf{s}'})\, p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) d\Omega'$$

Radiative transfer equation

General energy conservation:

$$\frac{1}{c}\frac{I(r,\hat{\mathbf{s}})}{\partial t} + \hat{\mathbf{s}}\cdot[\nabla I(r,\hat{\mathbf{s}})] + \alpha I(r,\hat{\mathbf{s}}) = 0$$

RTE:

$$\frac{1}{c}\frac{\partial I(\mathbf{r},\hat{\mathbf{s}})}{\partial t} + \underbrace{\hat{\mathbf{s}}\cdot[\nabla I(\mathbf{r},\hat{\mathbf{s}})]} + \mu_t I(\mathbf{r},\hat{\mathbf{s}}) - \mu_t \int_{4\pi} I(\mathbf{r},\widehat{\mathbf{s}'})\, p(\widehat{\mathbf{s}'},\hat{\mathbf{s}}) d\Omega' = \epsilon(r,\hat{\mathbf{s}})$$

$$\hat{\mathbf{s}}\cdot[\nabla I(\mathbf{r},\hat{\mathbf{s}})] = \nabla\cdot[I(\mathbf{r},\hat{\mathbf{s}})\hat{\mathbf{s}}] = \frac{\partial I}{\partial s}$$

Summary of assumptions

- Far-field approximation: phase function is calculated/measured in the far-field, Poynting's theorem in its far-field form
- There is a middle scale δV:
  - δV larger than the wavelength
  - δV contains many scatterers – statistically equivalent optical properties in the medium
  - δV small enough to consider the Poynting vector there as almost constant
  - Light scattered from δV is coupled to other volumes (light cannot scatter back into δV)
- Incoherence: summation of energy and not fields
- Polarization effects neglected

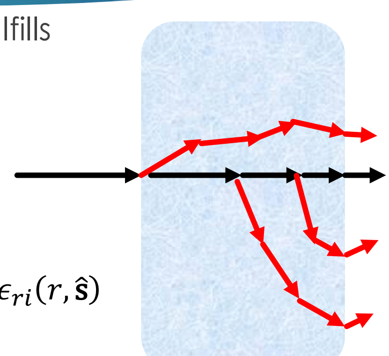
Reduced incident intensity

- The reduced incident intensity fulfills

$$\frac{dI_{ri}(\mathbf{r},\hat{\mathbf{s}})}{ds} = -\mu_t I_{ri}(\mathbf{r},\hat{\mathbf{s}})$$

- The total intensity is $I = I_{ri} + I_d$

$$\hat{\mathbf{s}} \cdot [\nabla I_d(\mathbf{r},\hat{\mathbf{s}})] + \mu_t I_d(\mathbf{r},\hat{\mathbf{s}}) - \mu_t \int_{4\pi} I_d(\mathbf{r},\hat{\mathbf{s}}')p(\hat{\mathbf{s}}',\hat{\mathbf{s}})d\Omega' = \epsilon(r,\hat{\mathbf{s}}) + \epsilon_{ri}(r,\hat{\mathbf{s}})$$



where $\epsilon_{ri}(r,\hat{\mathbf{s}}) = \mu_t \int_{4\pi} I_{ri}(\mathbf{r},\hat{\mathbf{s}}')p(\hat{\mathbf{s}}',\hat{\mathbf{s}})d\Omega'$

Integral form

$$\frac{dI(\mathbf{r},\hat{\mathbf{s}})}{ds}+\mu_t I(\mathbf{r},\hat{\mathbf{s}})-\mu_t\int_{4\pi} I(\mathbf{r},\widehat{\mathbf{s}'})p(\widehat{\mathbf{s}'},\hat{\mathbf{s}})d\Omega'=0$$

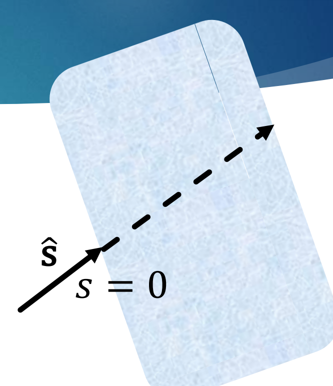
This is a first order differential equation of the form:

$$\frac{dy}{dx}+Py=Q(x)$$

The solution:

$$I(\mathbf{r},\hat{\mathbf{s}})=ce^{-\mu_t s}+ce^{-\mu_t s}\int Q\,(\widehat{\mathbf{s}'})e^{\mu_t s'}ds'$$

Integral form



$$I(\mathbf{r},\hat{\mathbf{s}})$$
$$=I_0e^{-\mu_t s}+\int_0^s e^{-\mu_t(s-s_1)}\left[\int_{4\pi} I(\mathbf{r}_1,\widehat{\mathbf{s}'})p(\widehat{\mathbf{s}'},\hat{\mathbf{s}})d\Omega'\right]ds_1$$

Analytical solution in the 1D case in steady state

$$\int_{4\pi} I(r,\widehat{\mathbf{s}'})\, p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) d\Omega'$$

$$d\Omega' = cos\theta d\theta d\varphi = d(cos\theta) d\varphi$$

Assuming isotropy in φ

$$\int_{4\pi} I(r,\widehat{\mathbf{s}'})\, p(\hat{\mathbf{s}},\hat{\mathbf{s}}_0) d\Omega' \rightarrow \int_{-1}^{1} I(x,\tau')\, p(\tau,\tau') d\tau$$

Where $\tau = cos\theta$

Analytical solution in the 1D case in steady state

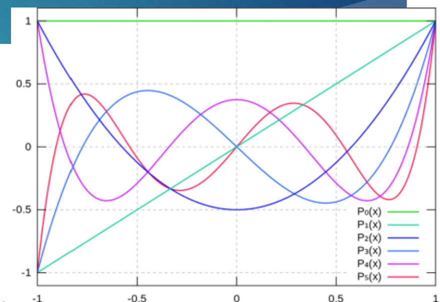
$$\tau \frac{\partial}{\partial x}\psi(x,\tau') + \mu_t \psi(x,\tau') = \mu_s \int_{-1}^{1} \psi(x,\tau')\, f(\tau,\tau') d\tau' + \frac{S(x)}{2}$$

- The standard approach: represent the problem in a new orthogonal basis
- For τ we choose the Legendre polynomials, which are orthogonal on [-1,1]

$$\psi(x,\tau) = \sum_{n=0}^{\infty} \frac{2n+1}{2} \phi_n(x) P_n(\tau) \qquad f(\tau,\tau') = \sum_{n=0}^{\infty} \frac{2n+1}{2} f_n P_n(\tau) P_n(\tau').$$

A. Liemert and A. Kienle, Phys. Rev. A **83**, 015804 (2011).

Legendre polynomials



n	$P_n(x)$
0	1
1	x
2	$\frac{1}{2}(3x^2-1)$
3	$\frac{1}{2}(5x^3-3x)$
4	$\frac{1}{8}(35x^4-30x^2+3)$
5	$\frac{1}{8}(63x^5-70x^3+15x)$
6	$\frac{1}{16}(231x^6-315x^4+105x^2-5)$
7	$\frac{1}{16}(429x^7-693x^5+315x^3-35x)$
8	$\frac{1}{128}(6435x^8-12012x^6+6930x^4-1260x^2+35)$
9	$\frac{1}{128}(12155x^9-25740x^7+18018x^5-4620x^3+315x)$
10	$\frac{1}{256}(46189x^{10}-109395x^8+90090x^6-30030x^4+3465x^2-63)$

Analytical solution in the 1D case in steady state

Resulting equation:

$$\frac{n+1}{2n+1}\frac{d}{dx}\phi_{n+1}(x)+\frac{n}{2n+1}\frac{d}{dx}\phi_{n-1}(x)+\mu_{an}\phi_n(x)=S(x)\delta_{n,0},$$

Fourier decomposition:

$$\phi_n(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\phi_n(k)e^{ikx}dk.$$

$$\frac{n+1}{2n+1}ik\phi_{n+1}(k)+\frac{n}{2n+1}ik\phi_{n-1}(k)+\mu_{an}\phi_n(k)=\delta_{n,0}.$$

$$\mu_{an}=\mu_a+(1-f_n)\mu_s$$

$$\mathbf{T}[\phi_0(k),\phi_1(k),\ldots,\phi_N(k)]^T=[1,0,\ldots,0]^T$$

Analytical solution in the 1D case in steady state

- To solve the problem, we need to invert the following matrix relation:

$$\mathbf{T} = \begin{pmatrix} \mu_a & ik & 0 & 0 & \cdots & 0 \\ ik & 3\mu_{a1} & i2k & 0 & \cdots & \vdots \\ 0 & i2k & \ddots & \ddots & \cdots & 0 \\ 0 & 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \cdots & \cdots & \ddots & \ddots & iNk \\ 0 & \cdots & 0 & 0 & iNk & (2N+1)\mu_{aN} \end{pmatrix}$$

- We cannot calculate each order separately
- The fact that (2n+1)μ_{an} increases with *n* makes the higher-order solutions smaller
- $\mu'_s \gg \mu_a$ leads to reduction in the higher-order modes

Numerical solution to RTE – the Monte Carlo approach

- Monte Carlo simulations: A stochastic approach to solving differential equations
- Example: Radioactive decay $\frac{dN}{dt}$=-λN or ΔN=-$\lambda N \Delta t$
- We stochastically calculate the time it takes for a single emission, and repeat the process until we get a set of times with all the emissions
- In radioactive decay, the probabilistic model is a physical model, but the Monte Carlo approach may be applied as a mathematical solution, even when there's not physical interpretation behind it

Monte Carlo solution to RTE

- We start with a "photon packet" with a weight *W=1*, position (x,y,z)=(0,0,0) and direction vector $\hat{\mathbf{n}} = (0,0,1)$ – Pencil beam
- We stochastically calculate the time until the photon packet experiences an interaction with the medium (absorption or scattering).
- For each interaction, we either scatter or absorb the photon
- We continue until *W* reaches zero

Time until interaction

- Based on the term $\frac{1}{\mathrm{c}}\frac{I(r,\hat{\mathbf{s}})}{\partial t} = -\mu_t I(r,\hat{\mathbf{s}})$ in the RTE, we write

  $P(t > t_1) = \exp(-c\mu_t t_1)$ and $P(t < t_1) = 1 - \exp(-c\mu_t t_1)$
- The probability density function (PDF) of t is accordingly

  $p(t) = c\mu_t \exp(-c\mu_t t)$
- We draw a value for the it takes the photon packet to interact by using the probability function
- We update the position of the photon packet:

  $x \leftarrow x + cn_x t_i; \quad y \leftarrow y + cn_x t_i; \quad z \leftarrow z + cn_x t_i$

Interaction site

- The photon packet experiences both absorption and scattering
- The weight of the photon packet changes due to absorption:

$$W \leftarrow W - \Delta W \text{ where } \Delta W = \frac{\mu_a}{\mu_t} W$$

- The photon packet receives a new direction, drawn using the phase function

Termination

- We cannot simply terminate the process when W is small. That would create a bias and we won't allow us to see the "far reaching" photons
- Termination without bias: once W reaches a threshold we perform a *Russian roulette*

$$W \leftarrow \begin{cases} mW & \text{with a probability of } 1/m \\ 0 & \text{otherwise} \end{cases}$$

Monte Carlo: Summary

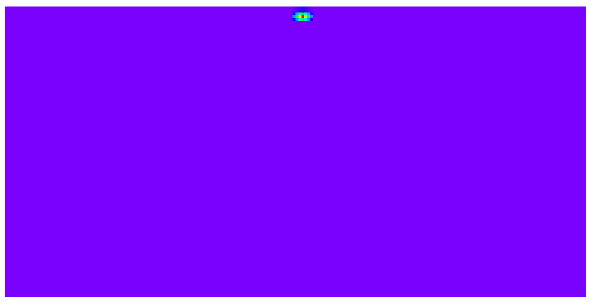
- The positions of the interaction sites are projected to the grid and their corresponding times and deposited energy ΔW are saved
- Numerous photon packets are launched into the medium and the results are averaged
- For each time interval $[\mathrm{t}, \mathrm{t} + \Delta]$, we can obtain a map of absorbed energy. If we divide by μ_a, we get the fluence rate

Generalization

- Spatial variations in refractive optical properties may be introduced (refractive indices, absorption, scattering).
- Versatile illumination profiles may be modelled (sum many pencil beam or perform convolution)
- GPU implementations enable fast calculations
- The reflection and transmission of a medium may be calculated

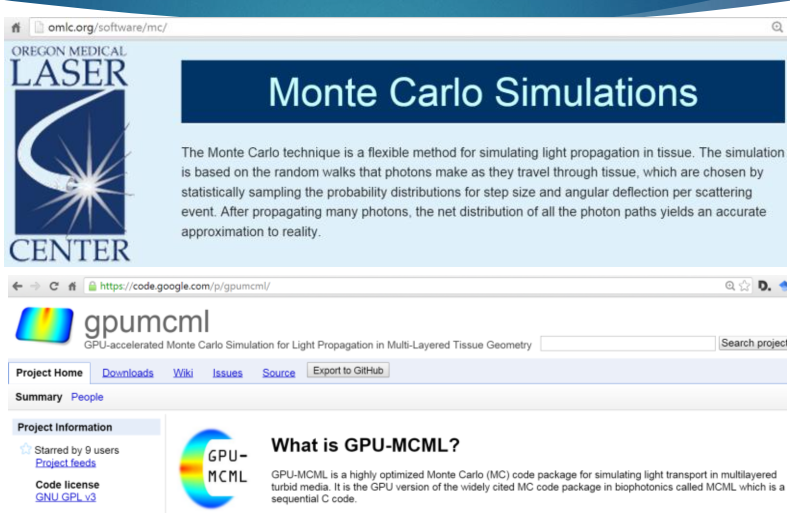
Example

Monte Carlo simulation of a pencil beam incident on a semi-infinite scattering medium:

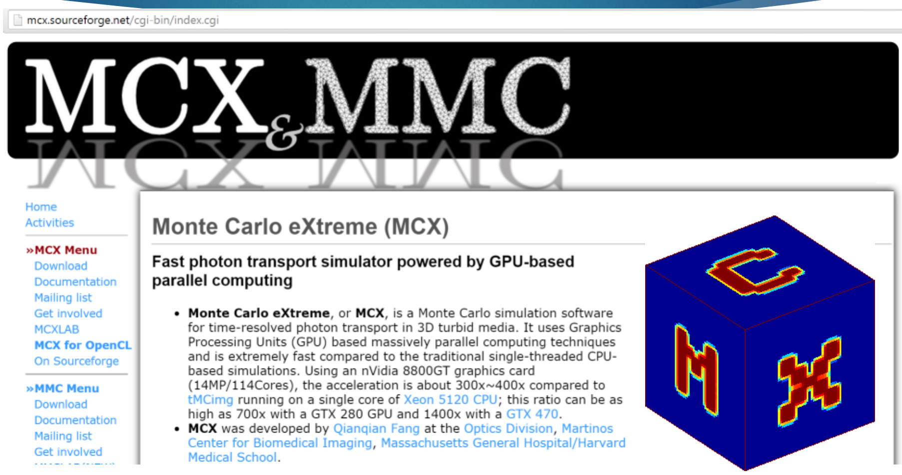


https://en.wikipedia.org/wiki/Monte_Carlo_method_for_photon_transport

Monte Carlo: Online recourses



Monte Carlo: Online recourses



Important to remember

The photon packets in Monte Carlo simulations do not have a physical meaning. They are not photons and do not represent any effect of quantum physics.